FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: PLANT GROWTH AND DEVELOPMENT	
Student Assessment Document	
Student Name	______________________________________
Class	______________________________________
Date	______________________________________

INSTRUCTIONS FOR STUDENTS

You have now completed all eight lessons on Plant Growth and Development. It is time to pull everything you have learned together into one Final Explanation that answers the driving question: "Why do seeds from the same garden grow and develop so differently, and what controls how a plant grows from a tiny seed into a tall tree or a flowering herb?" Use your Summary Table, your lab and field observations, the models you have built and revised across lessons, and any evidence from experiments you conducted throughout the unit. Your Final Explanation must directly connect back to the Kisumu farmer's garden — the maize and bean seedlings, the old mango and avocado trees, and the seeds that refused to germinate — and use that phenomenon as your anchor throughout. Include specific scientific vocabulary you have learned (e.g., epigeal, hypogeal, germination, meristems, auxin, gibberellin, primary and secondary growth, dormancy, tropism). Where relevant, draw on Kenyan examples such as crops, trees, and farming contexts you have explored. Your answer must be at least 300 words and should be written in clear, well-organised paragraphs — one section at a time. Do not simply list facts; explain the science behind what you observed and connect cause to effect throughout.

SECTION 1: GERMINATION — WHY SEEDS SPROUT (OR DON'T)

Explain what germination is and what conditions are required for it to occur. Use the Kisumu farmer's phenomenon to explain why some seeds in the packet that had been stored for two seasons failed to germinate even when given water and warmth. What is seed dormancy, and what internal and external factors must be overcome before a seed can sprout? Use specific evidence from your lessons and experiments.	Germination is the process by which a seed breaks out of its dormant state and begins to grow into a new plant. For germination to happen, three essential external conditions must be met: the seed needs adequate water (imbibition), the right temperature range, and sufficient oxygen for cellular respiration. When the Kisumu farmer watered her garden equally, she provided water and warmth — yet some seeds from the two-season-old packet still refused to germinate. This is because those seeds were no longer viable. Over time, the embryo inside a stored seed can die or become too damaged to resume growth, especially if the seeds were stored in a warm, humid environment like many Kenyan homes. Even if the outer conditions are perfect, a dead or weakened embryo cannot respond. Seed dormancy is a survival strategy that prevents a seed from germinating at the wrong time — for example, during a dry spell that would kill a young seedling. Dormancy can be physical (a hard seed coat that blocks water entry, as seen in some bean varieties) or physiological (internal hormone imbalances, particularly low gibberellin levels, that prevent the embryo from activating). For dormancy to break, the seed may need a period of cold temperatures, scarification of the seed coat, or the right balance of plant hormones. In the farmer's case, the old seeds likely suffered from loss of embryo viability rather than simple dormancy — a reminder that seed storage conditions in Kisumu's humid climate matter enormously. Farmers across the Rift Valley are often advised to store seeds in cool, dry, airtight containers to preserve viability between growing seasons.

SECTION 2: EPIGEAL VS. HYPOGEAL GERMINATION — WHY MAIZE AND BEANS EMERGE DIFFERENTLY	
The farmer noticed that bean seedlings pushed their cotyledons above the soil and formed an arch shape, while maize seedlings sent up a single green spike with the seed staying underground. Explain the difference between epigeal and hypogeal germination. What structures are involved in each type, and what causes the differences in emergence pattern? Why does it matter whether the cotyledons appear above or below ground?	The striking difference between how the bean and maize seedlings emerged from the Kisumu garden bed is explained by two distinct types of germination: epigeal and hypogeal. In epigeal germination — shown by the bean — the hypocotyl (the section of stem below the cotyledons) elongates rapidly and pushes the cotyledons up through the soil. The seedling forms a characteristic arch or hook shape, which protects the delicate growing tip as it pushes through the soil particles. Once above ground and exposed to light, the hypocotyl straightens, and the cotyledons unfurl and begin to photosynthesise. The cotyledons in beans are thick and fleshy because they are packed with stored food (mainly starch and protein) that fuels early growth — but they also briefly function as the plant's first photosynthetic leaves. Maize, on the other hand, undergoes hypogeal germination. The seed remains underground while the epicotyl (the shoot above the cotyledon) elongates and pushes upward, protected by a special sheath called the coleoptile — the single green spike the farmer observed. The single cotyledon (scutellum) of the maize grain stays below ground and acts as an absorptive organ, transferring stored nutrients from the endosperm to the growing embryo rather than rising above the soil. This is why maize seedlings look so different from bean seedlings even though both were planted in the same soil on the same day. The type of germination is genetically determined — it is encoded in the DNA of each species, not by the environment. This explains why the same soil, water, and temperature produce two completely different emergence patterns: the instructions come from inside the seed itself.

SECTION 3: PRIMARY GROWTH — HOW PLANTS GET TALLER AND LONGER	
Explain how plants grow in height and how roots extend deeper into the soil. What are meristems, and where are they found? What role do plant hormones — particularly auxin and gibberellin — play in controlling primary growth? Connect your explanation to the bean and maize plants in the farmer's garden growing taller over the weeks after germination.	After the bean and maize seedlings emerged from the soil in the Kisumu garden, both plants began growing rapidly upward and downward — this is primary growth. Primary growth is the increase in the length of a plant, and it occurs at special regions called meristems. Meristems are zones of actively dividing cells found at the tips of shoots (apical meristems) and the tips of roots (root apical meristems). In both the maize and bean plants, cells produced at the shoot apical meristem divide, elongate, and differentiate into all the specialised tissues of the stem and leaves, allowing the plant to grow taller week by week. Plant hormones play a critical role in directing this growth. Auxin, produced in the shoot tip, travels downward and stimulates cell elongation in the stem — the more auxin present, the faster the cells below the tip stretch and lengthen. This is why cutting off the tip of a plant (as farmers sometimes do when pruning) can cause side branches to grow instead, because auxin normally suppresses lateral buds. Gibberellins are another group of hormones that promote stem elongation and also play a key role in breaking seed dormancy and triggering germination. When a maize or bean seed absorbs water, gibberellin levels rise, activating enzymes that digest stored starch and release energy for the growing embryo. As the weeks passed in the Kisumu garden, the interplay of auxin, gibberellin, and other signals caused the bean plants to branch outward — because beans have a branching growth pattern encoded in their genetics — while maize grew as a single, tall, unbranched stem. Same hormones, same garden, but different genetic programmes producing different plant shapes.

SECTION 4: SECONDARY GROWTH — HOW PLANTS BECOME WOODY	
The farmer's garden also contained mango and avocado trees that had been planted years earlier. These trees had thick, woody trunks and wide	While the maize and bean plants in the Kisumu garden remained thin and flexible throughout their lives, the mango and avocado trees standing nearby had developed thick, hard, woody trunks over many years. This difference is explained by secondary growth — the increase in the girth (thickness) of a plant's stem and roots. Secondary growth is driven by two types of lateral meristems: the vascular cambium and the cork cambium. The vascular cambium is a ring of dividing cells found between the xylem and phloem; it produces new xylem cells (wood) toward the inside of the stem and new phloem cells toward the outside, causing the

canopies — very different from the thin, soft stems of the bean and maize plants nearby. Explain secondary growth: what causes it, what tissues are involved (vascular cambium, cork cambium), and why some plants (like maize and beans) do not undergo secondary growth while others (like mango and avocado trees) do. Connect this to the classification of plants as herbaceous or woody.	trunk to thicken year after year. In Kenya's tropical climate, mango and avocado trees grow year-round, but the rate of cambium activity can vary with rainfall seasons, which is why you can sometimes see growth rings in cross-sections of tree trunks. The cork cambium produces the tough, protective bark that covers the outside of woody stems, replacing the original epidermis as the trunk expands. Maize and beans, however, are monocots and herbaceous dicots respectively — neither undergoes significant secondary growth. Maize is a monocot and lacks a vascular cambium entirely; its vascular bundles are scattered throughout the stem rather than arranged in a ring, so there is no cambial layer to produce wood. Beans are herbaceous dicots that complete their life cycle within one growing season, so they never have time to develop significant secondary growth before they flower, set seed, and die. This is why the farmer can pull up a bean plant by hand but would need a chainsaw to fell the mango tree. Secondary growth is not just about age — it is about having the right genetic toolkit to activate a vascular cambium and sustain it over many years.

SECTION 5: BRINGING IT ALL TOGETHER — ANSWERING THE DRIVING QUESTION	
Now connect everything you have learned to answer the driving question fully: Why do seeds from the same garden grow and develop so differently, and what controls how a plant grows from a tiny seed into a tall tree or a flowering herb? In your answer, explain the roles of genetics, hormones, environment, and plant structure. Reflect on why understanding plant growth matters for farmers in Kenya — think about food security, crop management, and the health of trees like mango and avocado that support Kenyan livelihoods.	The Kisumu farmer's garden revealed one of the most important ideas in biology: that even when the environment is identical — same soil, same water, same sunlight — living things can grow in completely different ways, because growth is controlled from the inside as much as from the outside. Seeds from the same garden grow differently because each species carries its own genetic instructions. Those instructions determine whether germination is epigeal or hypogeal, whether the plant will be herbaceous or woody, how tall it will grow, and when it will flower. The maize and beans followed completely different developmental programmes encoded in their DNA, which is why their seedlings looked so unlike each other from the very first day of emergence. But genetics alone does not tell the whole story. Plant hormones — auxin, gibberellin, cytokinin, ethylene, and abscisic acid — act as the chemical messengers that translate genetic instructions into physical growth. They control when a seed breaks dormancy, how fast a stem elongates, which direction roots and shoots grow (tropisms), when flowers form, and when leaves drop. The environment modulates these hormones: light, temperature, water availability, and gravity all influence hormone levels and distribution. A seed stored poorly loses viability; a seedling shaded by a taller neighbour produces more auxin and stretches toward the light; a mango tree in Kisumu's warm, rainy climate grows its cambium actively and builds a wide, strong trunk over decades. Understanding plant growth is not just a scientific exercise — it has real consequences for Kenyan farmers and food security. Knowing why seeds lose viability helps farmers store maize and bean seeds correctly between seasons. Understanding dormancy helps agricultural officers advise on the best planting times. Knowing how auxin works helps farmers prune mango and avocado trees to increase fruit yield. And understanding secondary growth helps communities protect old trees, whose thick trunks represent years of growth that cannot be quickly replaced. The science of plant growth connects the laboratory to the farm, and from the farm to the plate — from a tiny seed in Kisumu's soil to ugali and beans on a family's dinner table.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Germination & Dormancy	Excellent (4): Accurately and fully explains the internal and external	Proficient (3): Correctly explains the main conditions for germination and defines	Developing (1-2): Identifies one or two conditions for germination but does not

	conditions required for germination (water, temperature, oxygen, viable embryo). Clearly defines seed dormancy (physical and physiological) and explains with specific evidence why the stored seeds in the phenomenon failed to germinate. Uses correct vocabulary (imbibition, viability, gibberellin, dormancy) consistently and accurately.	dormancy. Connects the failed germination to seed viability or dormancy with partial evidence. Uses most key vocabulary correctly but may miss one or two terms or slight nuances.	clearly explain dormancy or why stored seeds fail. Little connection to the phenomenon. Key vocabulary is missing or used incorrectly.
Epigeal vs. Hypogeal Germination	Excellent (4): Clearly and accurately distinguishes epigeal germination (bean — hypocotyl elongates, cotyledons rise above soil, arch shape) from hypogeal germination (maize — coleoptile rises, cotyledon/scutellum stays underground). Explains the function of the structures involved and correctly states that the type of germination is genetically determined, not environmentally controlled. Directly references the Kisumu farmer's observation.	Proficient (3): Correctly identifies epigeal and hypogeal germination and links each to the correct plant. Describes the key structural difference (cotyledons above vs. below ground) and mentions genetic determination. May not fully explain all structures (e.g., coleoptile, scutellum, hypocotyl) or their functions.	Developing (1-2): Shows some awareness that maize and beans emerge differently but cannot accurately explain epigeal vs. hypogeal germination or name the structures involved. May confuse which plant undergoes which type, or attribute the difference to environmental rather than genetic causes.
Primary Growth & Plant Hormones	Excellent (4): Accurately explains primary growth as increase in length driven by apical meristems. Correctly describes the roles of auxin (cell elongation, apical dominance) and gibberellin (stem elongation, dormancy breaking) with specific, accurate detail. Connects hormone action to observations of the maize and bean plants growing taller in the farmer's garden. Uses all key vocabulary correctly (meristem, auxin, gibberellin, apical dominance, cell elongation).	Proficient (3): Correctly explains that primary growth occurs at meristems and involves hormones. Describes auxin and gibberellin roles with reasonable accuracy. Makes a connection to the phenomenon but may lack specific detail or miss one hormone function. Key vocabulary is mostly used correctly.	Developing (1-2): Mentions that plants grow taller but does not accurately explain the role of meristems or hormones. May name auxin or gibberellin without correctly describing what they do. Little or no connection to the phenomenon or the Kisumu plants.
Secondary Growth & Herbaceous vs. Woody Plants	Excellent (4): Accurately explains secondary growth as increase in girth driven by vascular cambium (producing xylem and phloem) and cork cambium (producing bark). Clearly explains why maize (monocot, no vascular cambium, scattered bundles) and beans (herbaceous, short lifecycle) do not undergo secondary growth, while mango and avocado trees do. Directly connects to the phenomenon (the thick trunks of	Proficient (3): Correctly explains that secondary growth causes thickening and involves the vascular cambium. Distinguishes herbaceous from woody plants and connects each to the phenomenon. May not fully explain the role of the cork cambium or give a complete reason why monocots lack secondary growth. Key vocabulary is mostly used correctly.	Developing (1-2): Shows some awareness that trees are thicker/woodier than bean or maize plants but cannot accurately explain secondary growth, the role of cambia, or why some plants are herbaceous. Key vocabulary is largely absent or incorrectly used.

	the old trees vs. the thin stems of maize and beans). Uses all key vocabulary correctly (vascular cambium, cork cambium, xylem, phloem, herbaceous, woody).		
Synthesis, Kenya Context & Communication	Excellent (4): The Final Explanation is well-organised, clearly written, and meets the 300-word minimum. All sections connect back to the Kisumu farmer's phenomenon in a meaningful, specific way. Kenyan examples (crops, trees, farming practices, food) are woven naturally into the explanation. The student demonstrates genuine understanding by explaining relationships between concepts (e.g., genetics → hormones → structure → growth outcome) rather than simply listing facts. Scientific vocabulary is used accurately and consistently throughout.	Proficient (3): The explanation is organised and meets the word count. Most sections connect to the phenomenon. Some Kenyan examples are used. The student shows reasonable understanding of how concepts relate, though some explanations may be surface-level or lack full development. Scientific vocabulary is mostly used correctly.	Developing (1-2): The explanation is disorganised, below the word count, or reads as a list of disconnected facts. Connection to the Kisumu phenomenon is weak or absent. Few or no Kenyan examples are used. Scientific vocabulary is largely missing, incorrect, or used without understanding.